Deep Learning Interview Questions

Section 1: Deep Learning Fundamentals

1. What is Deep Learning, and how does it differ from traditional Machine Learning?
2. What is an artificial neural network (ANN)?
3. What is a biological neuron, and how is it related to an artificial neuron?
4. What are the main components of an artificial neuron?
5. What are input, hidden and output layers in a neural network?
6. What are weights and biases in a neural network?
7. What is a forward pass in a neural network?
8. What is a loss function, and why is it necessary?
9. What is the difference between an epoch, a batch and an iteration?
10. What are the advantages and limitations of Deep Learning compared with traditional ML algorithms?

Section 2: Neural Networks and Activation Functions

11. What is an activation function, and why is it necessary?
12. What happens if a neural network uses no nonlinear activation functions?
13. Explain the sigmoid activation function and its limitations.
14. What is the tanh activation function, and how does it differ from sigmoid?
15. What is the ReLU activation function, and why is it widely used?
16. What is the dying ReLU problem?
17. What are Leaky ReLU, ELU and GELU?
18. What is the softmax function, and when is it used?
19. How do you choose an activation function for the hidden and output layers?
20. What is the difference between a single-layer perceptron and a multilayer perceptron (MLP)?

Section 3: Forward Propagation and Backpropagation

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21. What is forward propagation, and how are predictions calculated?
22. What is backpropagation, and why is it important?
23. How does the chain rule help calculate gradients in neural networks?
24. What is a gradient in the context of Deep Learning?
25. How are weights and biases updated during training?
26. What is the difference between a loss function and an optimisation algorithm?
27. What is the vanishing gradient problem?
28. What is the exploding gradient problem?
29. How can gradient clipping help stabilise training?
30. How would you calculate the output of a simple neural network with given weights, biases and activation functions?

Section 4: Loss Functions and Optimisation

31. What is Mean Squared Error (MSE), and when is it used?
32. What is Mean Absolute Error (MAE), and how does it differ from MSE?
33. What is binary cross-entropy loss?
34. What is categorical cross-entropy, and when is sparse categorical cross-entropy preferable?
35. What is gradient descent, and how does it minimise a loss function?
36. What is the difference between batch, stochastic and mini-batch gradient descent?
37. What is the learning rate, and what happens when it is too high or too low?
38. How does the Adam optimiser work, and how does it differ from SGD?
39. What are Momentum and RMSprop, and how do they improve optimisation?
40. What is a learning-rate scheduler, and why might you use one?

Section 5: Training, Validation and Regularisation

41. What is overfitting in a neural network, and how can you detect it?
42. What is underfitting, and how can you address it?
43. What is the bias–variance trade-off in Deep Learning?
44. What is dropout, and how does it reduce overfitting?
45. What is batch normalisation, and why is it useful?

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46. What is layer normalisation, and how does it differ from batch normalisation?
47. What is L1 and L2 regularisation in neural networks?
48. What is early stopping, and how does it work?
49. How do training and validation loss curves help diagnose model performance?
50. A model achieves excellent training accuracy but poor validation accuracy. How would you investigate and improve it?

Section 6: Convolutional Neural Networks

51. What is a Convolutional Neural Network (CNN), and why is it useful for image processing?
52. What is a convolution operation in a CNN?
53. What are filters or kernels in a convolutional layer?
54. What are stride and padding, and how do they affect output dimensions?
55. What is a feature map?
56. What is pooling, and what are max pooling and average pooling?
57. What is the difference between convolutional and fully connected layers?
58. What is the receptive field of a CNN?
59. What is the difference between a standard convolution and a depthwise-separable convolution?
60. How would you design a CNN to classify images into multiple categories?

Section 7: CNN Architectures and Computer Vision

61. What is LeNet, and what was its contribution to computer vision?
62. What was the significance of AlexNet in Deep Learning?
63. What is the difference between VGG and Inception architectures?
64. What is a residual connection, and how does ResNet use it?
65. How does DenseNet differ from ResNet?
66. What is transfer learning, and why is it useful for image classification?
67. What is the difference between feature extraction and fine-tuning in transfer learning?

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68. What is data augmentation, and which augmentation techniques are useful for images?
69. What is the difference between image classification, object detection and image segmentation?
70. How would you handle a small, imbalanced image dataset when training a CNN?

Section 8: Recurrent Neural Networks (RNNs)

71. What is a Recurrent Neural Network (RNN), and what types of problems is it designed to solve?
72. How does an RNN maintain information from previous time steps?
73. What is a hidden state in an RNN?
74. What is Backpropagation Through Time (BPTT)?
75. Why do standard RNNs struggle with long-term dependencies?
76. How do vanishing and exploding gradients affect RNN training?
77. What is a Long Short-Term Memory (LSTM) network?
78. What are the input, forget and output gates in an LSTM?
79. What is a Gated Recurrent Unit (GRU), and how does it differ from LSTM?
80. When would you choose an RNN, LSTM or GRU for a sequence prediction task?

Section 9: Natural Language Processing with Deep Learning

81. How is Deep Learning used in Natural Language Processing (NLP)?
82. What is tokenisation, and why is it important for neural language models?
83. What is a word embedding?
84. What is the difference between one-hot encoding and dense word embeddings?
85. How do Word2Vec and GloVe generate word representations?
86. What is the difference between CBOW and Skip-gram in Word2Vec?
87. What are the limitations of static word embeddings?
88. What is an encoder–decoder architecture, and how is it used in sequence-to-sequence tasks?

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89. What is the attention mechanism, and why was it introduced?
90. How would you build a sentiment classification model using Deep Learning?

Section 10: Transformers and Attention

91. What is a Transformer architecture, and how does it differ from an RNN?
92. What is self-attention, and how does it work?
93. What are query, key and value vectors in attention?
94. How is the scaled dot-product attention operation calculated?
95. What is multi-head attention, and what advantages does it provide?
96. Why do Transformers need positional encodings or another mechanism to represent token positions?
97. What is the difference between an encoder-only, decoder-only and encoder–decoder Transformer?
98. What are BERT and GPT, and how do their training objectives and typical uses differ?
99. What is causal masking, and why is it important for autoregressive language models?
100. What are the computational and memory limitations of standard self-attention for long sequences?

Section 11: Advanced Architectures and Generative Models

101. What is an autoencoder, and what are its encoder, latent representation and decoder?
102. What is a Variational Autoencoder (VAE), and how does it differ from a standard autoencoder?
103. What is a Generative Adversarial Network (GAN)?
104. What are the generator and discriminator in a GAN?
105. What is mode collapse in GAN training?
106. What is a diffusion model, and how does it generate new data?
107. What is the difference between discriminative and generative models?
108. What is self-supervised learning, and how does it differ from supervised learning?

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109. What is contrastive learning, and how are positive and negative pairs used?
110. What is knowledge distillation, and how can it create a smaller, more efficient model?

Section 12: Advanced Training Techniques

111. What is weight initialisation, and why does it matter?
112. What are Xavier and He initialisation, and when are they appropriate?
113. What is mixed-precision training, and how can it reduce memory usage?
114. What is gradient accumulation, and when is it useful?
115. What is gradient checkpointing, and what trade-off does it introduce?
116. What is distributed Deep Learning training?
117. What is the difference between data parallelism and model parallelism?
118. What is curriculum learning, and when might it improve training?
119. What is hyperparameter optimisation in Deep Learning, and which parameters are commonly tuned?
120. How would you debug a neural network whose loss becomes `NaN` during training?

Section 13: TensorFlow, Keras and PyTorch

121. What is TensorFlow, and what role does Keras play in its ecosystem?
122. What is PyTorch, and what are tensors in PyTorch?
123. What is the difference between TensorFlow/Keras and PyTorch?
124. What is the difference between a tensor and a NumPy array?
125. How do you define a neural network using the Keras Sequential API?
126. What are the roles of `compile()`, `fit()`, `evaluate()` and `predict()` in Keras?
127. What is an automatic differentiation system, and how does PyTorch's autograd work?
128. What is the difference between `model.train()` and `model.eval()` in PyTorch?
129. How do you save and load model weights and complete model checkpoints?

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130. How would you use a GPU to train a Deep Learning model, and how would you verify that computation is actually running on it?

Section 14: Deployment, Optimisation and MLOps

131. What is model inference, and how does it differ from model training?
132. What is model quantisation, and how can it reduce model size and inference cost?
133. What is pruning in neural networks, and how does it affect performance?
134. What is knowledge distillation, and when is it useful for deployment?
135. What is ONNX, and how can it support model portability?
136. What is the difference between batch inference and online inference for Deep Learning applications?
137. How would you deploy a trained neural network through a REST API using FastAPI or Flask?
138. How would you monitor latency, throughput, memory consumption and prediction quality in a deployed model?
139. What is model drift in a deployed Deep Learning system, and how would you detect it?
140. How would you optimise a Deep Learning model for deployment on a device with limited memory and processing power?

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